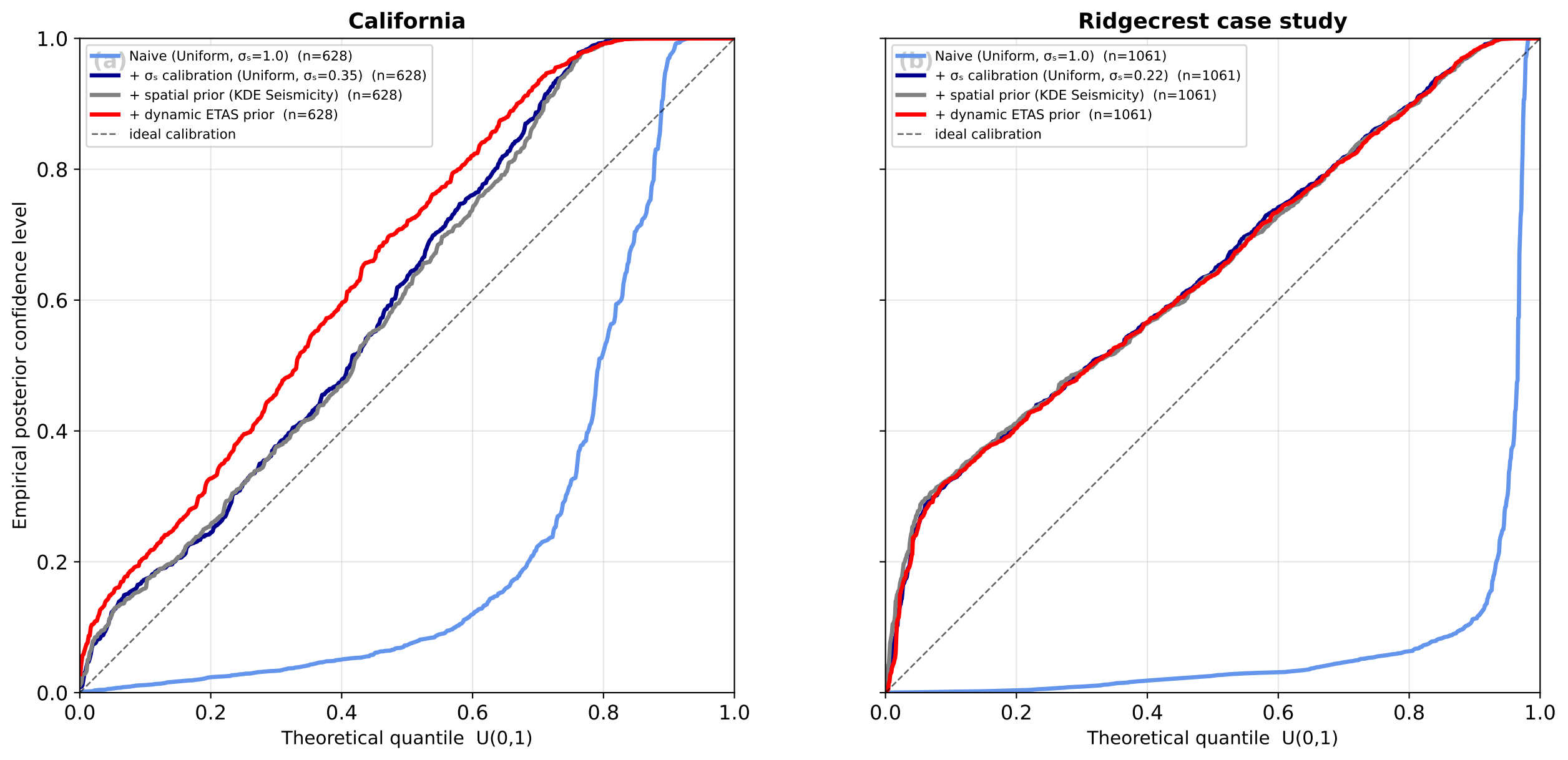

California
(a)
Naive (Uniform, σs=1.0) (n=628)
+ σs calibration (Uniform, σs=0.35) (n=628)
+ spatial prior (KDE Seismicity) (n=628)
+ dynamic ETAS prior (n=628)
ideal calibration
Empirical posterior confidence level
Theoretical quantile U(0,1)
0.0
0.2
0.4
0.6
0.8
1.0
Ridgecrest case study
(b)
Naive (Uniform, σs=1.0) (n=1061)
+ σs calibration (Uniform, σs=0.22) (n=1061)
+ spatial prior (KDE Seismicity) (n=1061)
+ dynamic ETAS prior (n=1061)
ideal calibration
Theoretical quantile U(0,1)